

The Chips Behind the Chatbots

AMD's MI400 Chips and Triple-Exaflop Helios Racks Ship the Same Week ChatGPT Passes TikTok in Downloads

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Yesterday this series covered the EU forcing Google to open Android's AI hooks to rivals -- a story about who gets to compete on top of the stack. Today's news sits one layer down. On July 22 AMD opened its Advancing AI 2026 conference in San Francisco with the full MI400 accelerator lineup and a Helios rack built to squeeze three AI exaflops into a single cabinet, its most direct pitch yet at Nvidia's data-center dominance. Twenty-four hours earlier, Google had shipped Gemini 3.6 Flash, a model that does more per token than its predecessor while costing less to run. And sometime in the last few weeks, without a keynote or a press release, ChatGPT quietly became the most-downloaded free app on both iOS and Android, ahead of TikTok. Three stories, one underlying fact: every dollar chipmakers strip out of an AI exaflop shows up, months later, as a cheaper token -- and every cheaper token shows up as more people opening the app. Today is about following that chain from silicon to download chart.

VIRAL / OPEN-SOURCE SPOTLIGHT

ChatGPT: The App That Out-Downloaded TikTok

As of this summer, OpenAI's ChatGPT has overtaken TikTok as the most-downloaded free app on both the iOS App Store and Google Play -- a milestone that would have sounded absurd three years ago, when ChatGPT was a research demo and TikTok was the platform every other app was chasing. No single feature explains it; it's the compounding effect of ChatGPT becoming the default way people search, draft, plan, and now delegate multi-step tasks to an agent, all inside one app that keeps getting cheaper to run as tiers like GPT-5.6's Luna and Gemini's Flash line drive inference costs down. Kling AI's dance-video trend and Nubia's agent phone (both covered earlier in this series) are viral the way a single feature goes viral; ChatGPT passing TikTok is viral the way a category itself tips -- when a chat interface with agentic reach becomes more habit-forming than short video, that's not a feature story, it's a platform-shift story.

3 AI exaflops per Helios rack	17% fewer output tokens, Gemini 3.6 Flash vs 3.5 Flash	42 state AGs investigating OpenAI	#1 ChatGPT's download rank, past TikTok
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1) AMD's answer to Nvidia: the full MI400 lineup and a three-exaflop rack

The MI400 series has been on AMD's roadmap since a spec sheet leaked back in June -- 432GB of HBM4 memory and 19.6 TB/s of bandwidth, aimed at hyperscalers looking to diversify away from Nvidia. Advancing AI 2026 is where that spec sheet became a shipping product line. AMD announced the full MI400 family, including the MI440X (an eight-GPU rack server paired with an Epyc Venice CPU,

built for training and inference at scale) and the MI430X, aimed at sovereign AI and hybrid computing deployments that need to keep data in-country. The headline system is Helios, a double-wide rack shipping in Q3 that packs up to three AI exaflops of FP4 performance into a single cabinet -- roughly a 10x jump over AMD's current generation for frontier-scale models.

The more important announcement wasn't a chip at all: ROCm 7, AMD's CUDA-equivalent software stack, now claims 3.5x the performance of ROCm 6 and support for every major training framework out of the box, alongside a new ROCm Enterprise AI tier and an AMD Developer Cloud for renting GPU time directly. AMD is also leaning on open standards -- UALink for GPU-to-GPU interconnect and UEC for networking -- instead of Nvidia's proprietary NVLink, betting that hyperscalers tired of single-vendor lock-in will pay a software-maturity tax now in exchange for negotiating leverage later. That's the real contest: not whose chip benchmarks higher this quarter, but whose ecosystem an enterprise is willing to commit a multi-year buildout to.

2) Gemini 3.6 Flash: the price war moves down a tier

One day before AMD's keynote, Google quietly shipped Gemini 3.6 Flash -- not a flagship announcement, more a routine SKU update, except that it uses 17% fewer output tokens than 3.5 Flash on the Artificial Analysis Index and up to 65% fewer tokens on some coding benchmarks, while improving multi-step orchestration and cutting the compile-failure rate on generated code. It's priced at \$1.50 per million input tokens and \$7.50 per million output tokens -- cheaper than the model it replaces, in the same week GPT-5.6's Luna tier and Grok 4.5 are competing on the exact same axis: more capability per token, not just more capability.

There's a pointed irony here. This is the same Google that, per yesterday's story, is under a binding EU order to open Android's AI hooks to competitors starting next year. Regulatory pressure at the platform layer hasn't slowed product execution at the model layer at all -- if anything, Google shipped a cheaper, more capable Flash tier in the same 48 hours the DMA order became public. Being forced to compete on someone else's turf and racing to win on your own turf are, it turns out, two separate contests that a company this size can run in parallel.

3) Why the chip race and the model race are actually one race

Follow the money and the causality runs in one direction. Helios-class racks lower AMD and Nvidia's cost per FLOP; labs like Google pass a slice of that saving through as a cheaper token price, which is exactly what Gemini 3.6 Flash's \$1.50/\$7.50 pricing represents. Cheaper tokens mean an app like ChatGPT can serve more agentic, multi-step requests to more free-tier users without the unit economics collapsing -- which is a real part of how it out-downloaded TikTok this summer. And a bigger download base is the justification hyperscalers cite when they sign the next multi-billion-dollar GPU order, which is what AMD spent July 22 trying to win a bigger share of. Chips, prices, and downloads aren't three trends running alongside each other; they're one flywheel, and today just happened to be the day all three turns landed in the same 24 hours.

MARKET SIGNAL

The same week ChatGPT's download count passed TikTok's, a coalition of 42 state attorneys general is actively investigating OpenAI -- subpoenas covering advertising, user engagement and retention

tactics, and how the app handles minors and seniors, filed days after OpenAI's confidential IPO paperwork became public. Scale and scrutiny are arriving on the same timeline, not a delayed one: the more downloads an AI app racks up, the faster it becomes a target for the exact regulatory attention that a pre-IPO company most wants to avoid. Popularity is not a safety signal, and a download chart topping TikTok says nothing about whether the 42 AGs find what they're looking for.

PRACTICAL TAKEAWAYS

1. Read chip announcements for the pricing signal, not the spec sheet

The exaflop count in an AMD or Nvidia keynote isn't the number that affects your budget -- the number that matters is what it does to \$/million-tokens six months later. Track API pricing trends, not GPU launches, if you're forecasting AI spend.

2. Re-check your mid-tier model choice now that Flash got cheaper again

If you picked a Flash- or Mini-class model for cost reasons even a month ago, Gemini 3.6 Flash's token-efficiency gains are worth a re-benchmark -- the price/performance frontier on small models is moving faster than most teams' procurement cycles.

3. Separate "most downloaded" from "most trusted" when you read AI adoption stats

ChatGPT beating TikTok in downloads and 42 state AGs investigating OpenAI are both true at the same time -- treat adoption milestones and regulatory scrutiny as two independent signals about the same company, not evidence against each other.

TOMORROW

AMD's Advancing AI conference runs through July 23 -- we'll cover what shipped on day two of the keynote and whether Nvidia answers before the week is out.

Topics covered so far

116 days in: frontier model families and their pricing tiers (GPT-5.6, Claude Sonnet 5, Gemini 3.5 Pro), the open-weight counter-punch (GLM-5.2, DeepSeek V4, Qwen 3.6, Kimi K3), physical AI leaving the demo stage (Figure, Boston Dynamics, Agility Robotics), OS-level agent phones (Nubia NaviX Ultra), the governance spectrum from voluntary charters (WAICO) to binding regulation with fines attached (the EU's DMA order on Google), and now, as of today, the compute layer underneath all of it -- AMD and Nvidia's fight over who builds the racks the entire race runs on.